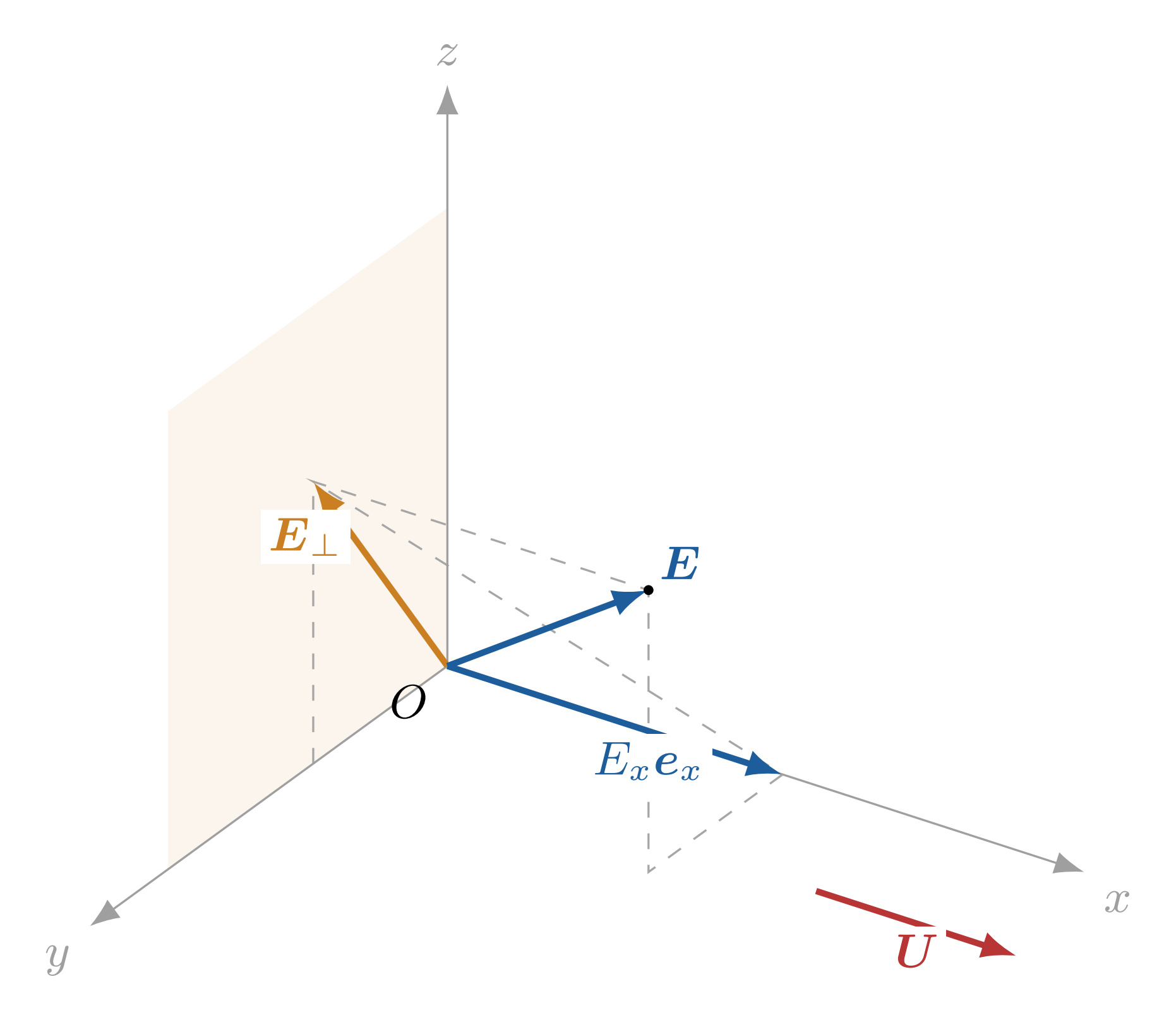
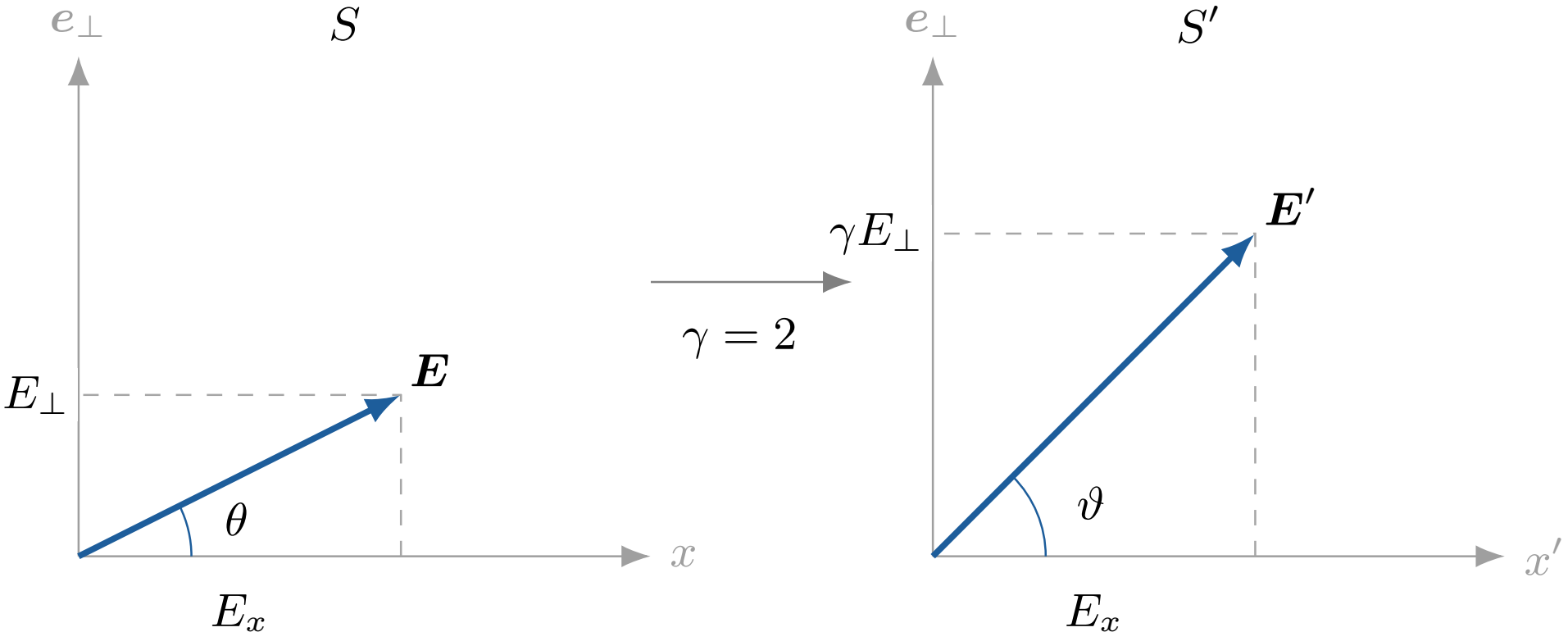
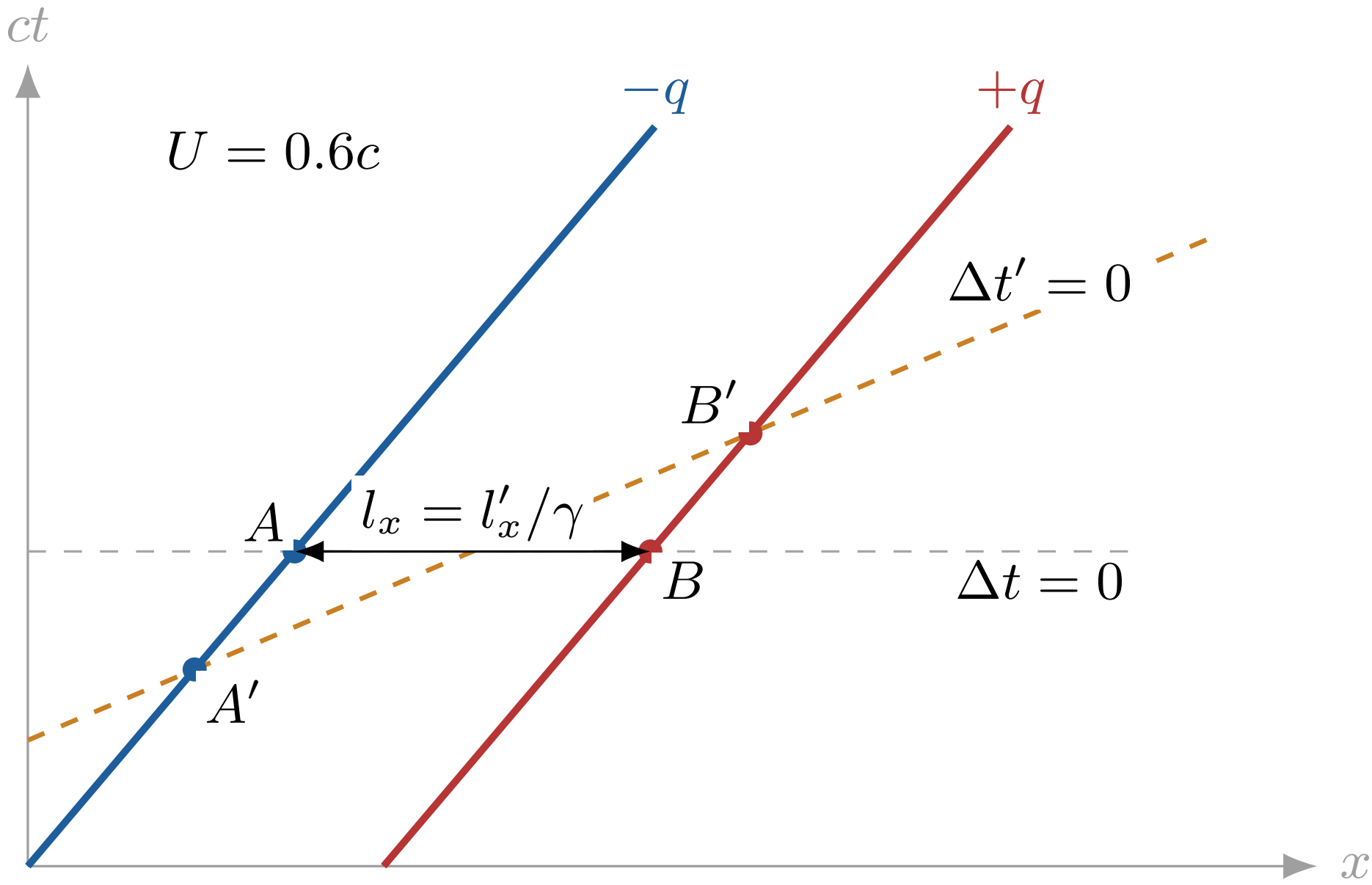
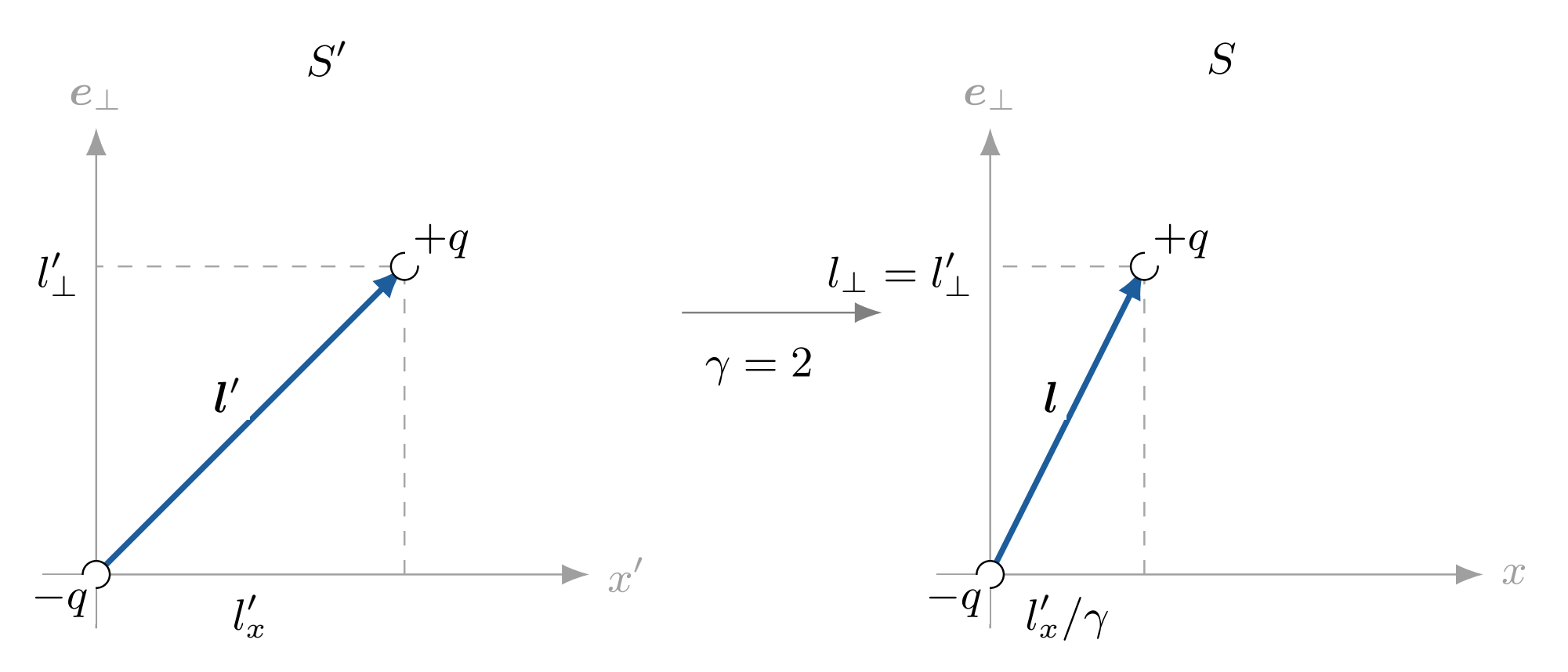
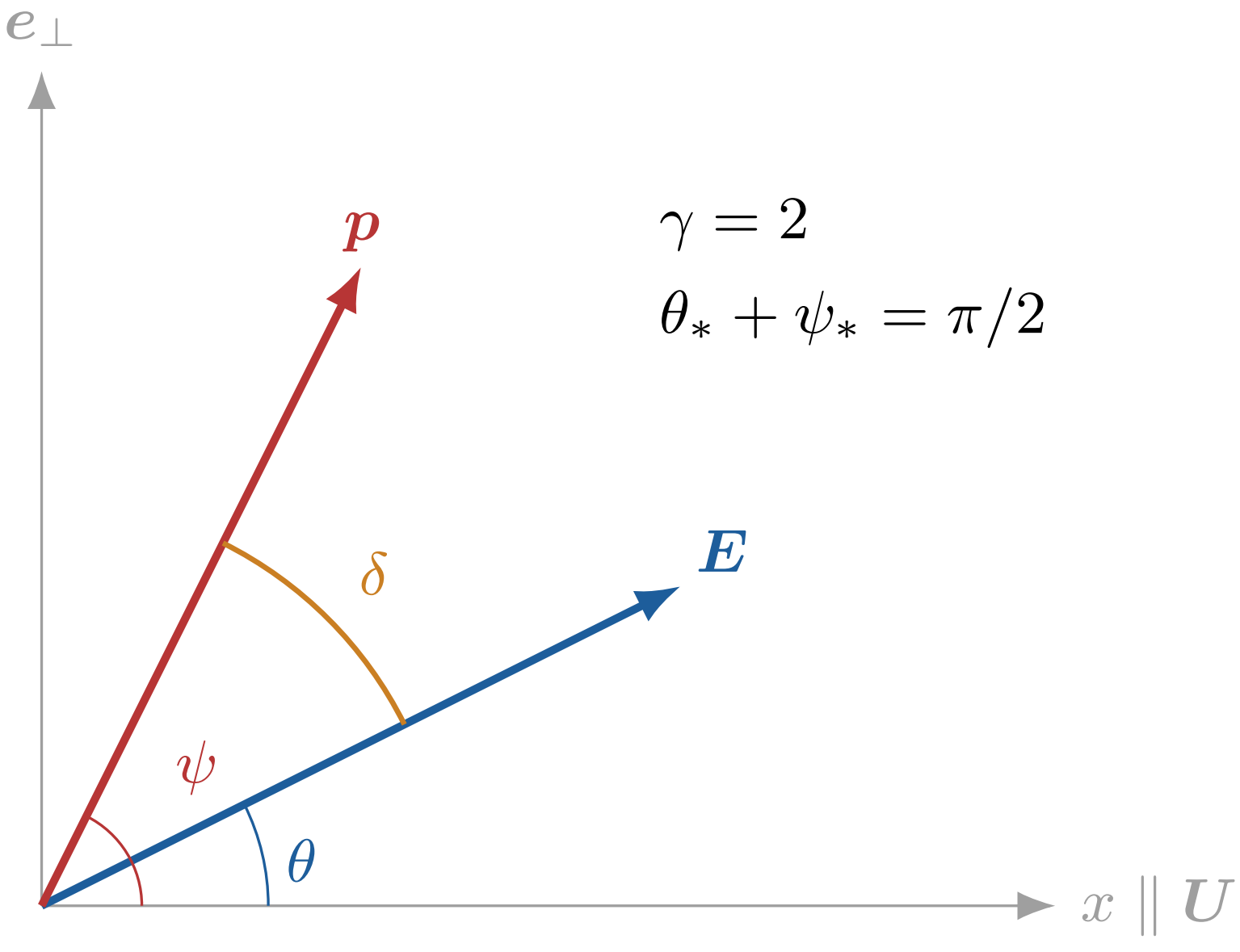
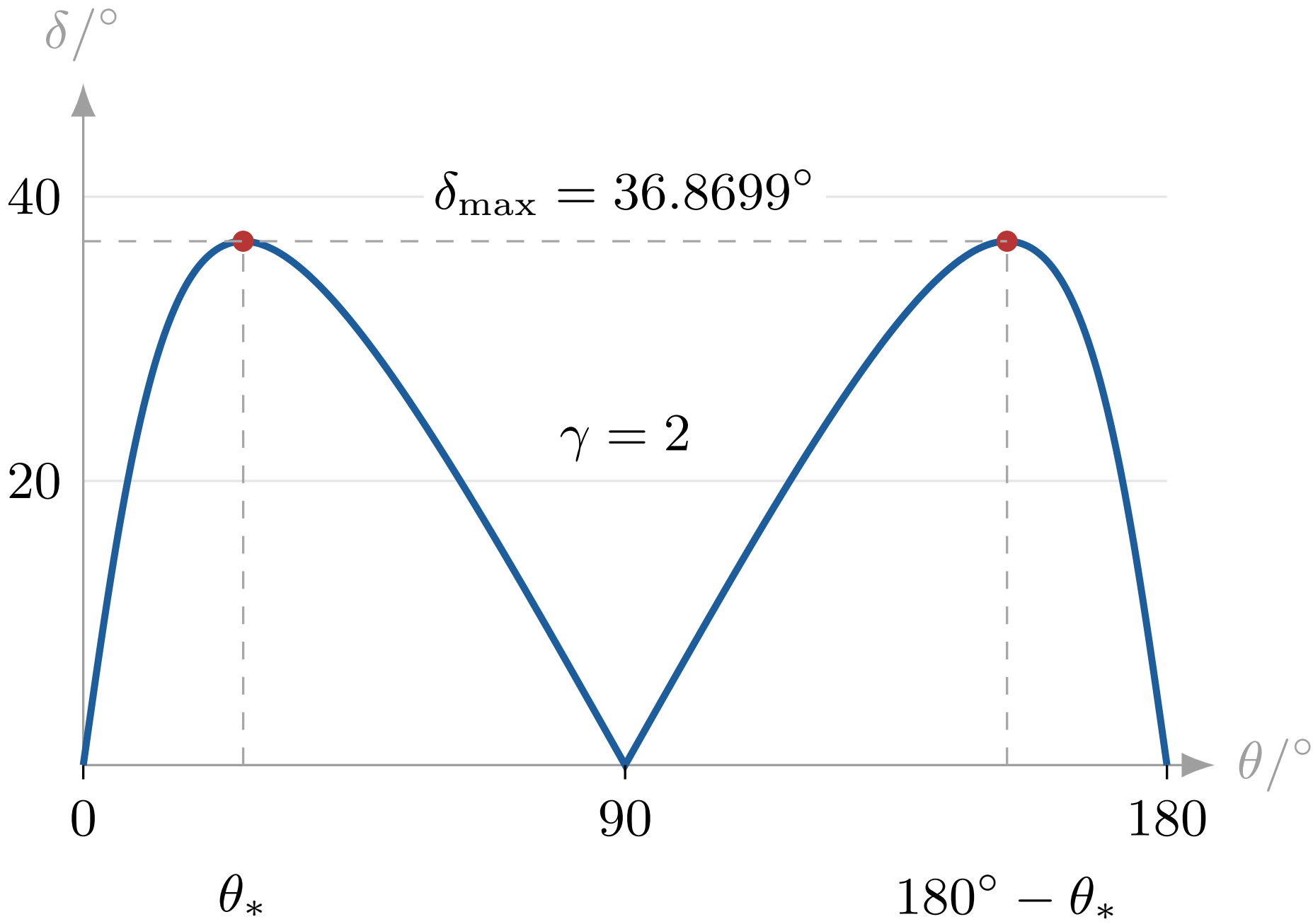

z
E⊥
E
O
Ex ex
x
y
U


$e_\perp$
S
$E_\perp$
$\boldsymbol{E}$
θ
x
E_x
$\gamma = 2$
$e_\perp$
S'
$\gamma E_\perp$
$\boldsymbol{E}'$
ϑ
x'
E_x


ct
−q
+q
U = 0.6c
Δt′ = 0
B′
A
l_x = l′_x/γ
B
Δt = 0
A′
x


S'
$e_\perp$
$+q$
$l'_\perp$
$\boldsymbol{l}'$
$-q$
l'_x
x'
$l_\perp = l'_\perp$
$\gamma = 2$
S
$e_\perp$
$+q$
$\boldsymbol{l}$
$-q$
l'_x/γ
x


$e_\perp$
$\boldsymbol{p}$
$\gamma = 2$
$\theta_* + \psi_* = \pi/2$
$\boldsymbol{E}$
δ
ψ
θ
$x \parallel \boldsymbol{U}$


$\delta/^\circ$
40
20
$\delta_{\max} = 36.8699^\circ$
$\gamma = 2$
$\theta/^\circ$
0
90
180
θ_*
$180^\circ - \theta_*$